

Sharp Support-Image Thresholds for Alternating Pencils

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Abstract

We prove an exact arity threshold for structured support-image observers on a natural infinite family of alternating pencils and associated class-two exponent- p groups. Let q be odd. For every monic irreducible polynomial $f \in \mathbb{F}_q[t]$ of degree $m \geq 3$, we construct a radical-free surjective alternating pencil $\beta_f : \Lambda^2 V_f \rightarrow W$ with $\dim_{\mathbb{F}_q} V_f = 2m$ and $\dim_{\mathbb{F}_q} W = 2$, whose Pfaffian is, up to scalar, the norm $N_{\mathbb{F}_{q^m}/\mathbb{F}_q}(a + b\theta)$, where θ is a root of f . Equivalently, β_f is the regular irreducible symplectic pencil attached to f . For $r \geq 1$, let $\text{CSI}_{\leq r}(\beta)$ denote the structured support-image profile through level r . We prove that $\text{CSI}_{\leq r}(\beta_f)$ is independent of f for every $r < m$, and that the residual projective observer symmetry is the full group $PGL_2(q)$ at every such level. At level m , the profile detects f up to the natural $PGL_2(q)$ -action. Thus, on irreducible regular pencils of degree m , the exact support-image separation depth is m . For $q = p$ prime, the associated Baer groups are directly indecomposable special class-two exponent- p groups of order p^{2m+2} , so the exact observer threshold is $(\log_p |G| - 2)/2$.

1 Introduction

Alternating maps

$$\beta : \Lambda^2 V \rightarrow W$$

control finite groups of nilpotency class two and exponent (p) . If (p) is odd, such a map defines a group

$$G_\beta = V \oplus W$$

with multiplication

$$(v, z)(u, w) \left(v + u, z + w + \frac{1}{2}\beta(v, u) \right).$$

The commutator is

$$[(v, z), (u, w)] = (0, \beta(v, u)).$$

Thus isomorphism questions for special class-two exponent- (p) groups are equivalent to pseudo-isometry questions for alternating maps.

This paper studies how much of an alternating map is visible to support-image observers. For

$$\mathbf{x} = (x_1, \dots, x_r) \in V^r,$$

define

$$A_{\mathbf{x}} : V \rightarrow W^r, \quad z \mapsto (\beta(x_1, z), \dots, \beta(x_r, z)).$$

The level-(r) support-image profile records the image subspaces

$$\text{im } A_{\mathbf{x}} \leq W^r$$

as $\mathbf{x}$ varies. It is a structured invariant of β , taken up to the natural action of $GL(W)$.

The central question is:

How large must r be before support-image data detects the map?

We answer this exactly for regular irreducible alternating pencils.

For each monic irreducible polynomial

$$f \in \mathbb{F}_q[t]$$

of degree (m), we construct a $2m$ -dimensional alternating pencil

$$\beta_f : \Lambda^2 V_f \rightarrow \mathbb{F}_q^2.$$

All nonzero scalar forms in the pencil are nondegenerate. Hence every first-order scalar-rank invariant sees the same object: a pencil in which every scalar member has full rank.

Nevertheless, the pencils are usually nonisomorphic. Their difference is encoded in the degree-(m) Pfaffian.

The main result says that support-image observers detect exactly this degree.

For every

$$r < m,$$

the complete structured support-image profile through level (r) is identical for all monic irreducible (f) of degree (m). Moreover, the residual projective symmetry visible to the observer is the full group

$$PGL_2(q).$$

At level (m), a single companion linearization of (f) becomes singular on β_f , and this detects (f) up to projective equivalence.

Thus the exact threshold is

$$r = m.$$

In group-theoretic terms, for $q = p$ prime, the associated groups have order

$$p^{2m+2},$$

so the exact threshold is

$$\frac{\log_p |G| - 2}{2}.$$

This gives a linear observer lower bound with a matching upper bound, on an exponentially large family of directly indecomposable special groups.

2 Structured support-image profiles

Let (q) be odd, and let (V, W) be finite-dimensional $\mathbb{F}_q$ -vector spaces.

An alternating map is a linear map

$$\beta : \Lambda^2 V \rightarrow W.$$

Equivalently, it is an alternating bilinear map

$$V \times V \rightarrow W.$$

For $x \in V$, define the contraction

$$A_x : V \rightarrow W, \quad z \mapsto \beta(x, z).$$

For an (r) -tuple

$$\mathbf{x} = (x_1, \dots, x_r) \in V^r,$$

define

$$A_{\mathbf{x}} : V \rightarrow W^r, \quad z \mapsto (A_{x_1} z, \dots, A_{x_r} z).$$

The level- (r) support-image profile is the function

$$T \leq W^r \mapsto E_r^\beta(T),$$

where

$$E_r^\beta(T) \# \{\mathbf{x} \in V^r : \text{im } A_{\mathbf{x}} = T\}.$$

The structured level- (r) support-image profile is this function on the full subspace lattice of W^r , considered together with the natural diagonal action of $GL(W)$.

It is often more convenient to use the equivalent rank transform.

Let

$$U \leq (W^r)^*.$$

Choose a basis

$$u_1, \dots, u_s$$

of (U) . Each u_j is a linear functional on W^r . Define

$$\Psi_U^\beta : V^r \rightarrow (V^*)^s$$

by

$$\Psi_U^\beta(\mathbf{x})(u_1 \circ A_{\mathbf{x}}, \dots, u_s \circ A_{\mathbf{x}}).$$

Set

$$\rho_r^\beta(U) \text{rank } \Psi_U^\beta.$$

This is independent of the chosen basis of (U) .

Proposition 2.1 (CSI/rank-transform equivalence). *For fixed (r) , the structured support-image profile*

$$T \mapsto E_r^\beta(T)$$

and the rank-transform function

$$U \mapsto \rho_r^\beta(U)$$

determine one another.

Proof. For

$$S \leq W^r,$$

define

$$F_r^\beta(S) \# \{\mathbf{x} \in V^r : \text{im } A_{\mathbf{x}} \leq S\}.$$

The condition

$$\text{im } A_{\mathbf{x}} \leq S$$

is equivalent to

$$u \circ A_{\mathbf{x}} = 0 \quad \text{for all } u \in S^\perp.$$

Thus

$$F_r^\beta(S) | \ker \Psi_{S^\perp}^\beta | q^{r \dim V - \rho_r^\beta(S^\perp)}. \quad (2.1)$$

On the other hand,

$$F_r^\beta(S) = \sum_{T \leq S} E_r^\beta(T). \quad (2.2)$$

Mobius inversion on the lattice of subspaces of W^r gives

$$E_r^\beta(T) \sum_{S \leq T} \mu(S, T) F_r^\beta(S). \quad (2.3)$$

Equations 2.1–2.3 recover E_r^β from ρ_r^β .

Conversely, E_r^β determines F_r^β by 2.2, and then 2.1 determines ρ_r^β .

□

□

3 Residual projective observer symmetry

Let

$$PGL(W) = GL(W)/\mathbb{F}_q^\times.$$

The group $PGL(W)$ acts on W , on W^r , and dually on $(W^r)^*$.

For

$$g \in PGL(W)$$

and

$$U \leq (W^r)^*,$$

write

$$gU$$

for the image of (U) under the induced dual action.

Definition 3.1 (Residual CSI symmetry). *The residual projective symmetry group of the structured CSI observer through level (r) is*

$$\Gamma_{\leq r}(\beta) \left\{ g \in PGL(W) : \rho_s^\beta(gU) = \rho_s^\beta(U) \text{ for all } 1 \leq s \leq r, U \leq (W^s)^* \right\}.$$

Equivalently, by Proposition 2.1, $\Gamma_{\leq r}(\beta)$ is the subgroup of $PGL(W)$ preserving all structured support-image profiles through level r .

If $(T, g) \in GL(V) \times GL(W)$ is a pseudo-isometry of β , i.e.

$$g\beta(x, y) = \beta(Tx, Ty),$$

then the projective class of (g) lies in $\Gamma_{\leq r}(\beta)$ for every (r) . Thus $\Gamma_{\leq r}(\beta)$ may be larger than the true projective automorphism group: it measures the false symmetry not yet broken by the observer.

4 Grassmann image profiles

We also use a parallel observer based on restrictions to subspaces.

For

$$r \geq 2$$

and

$$T \leq W,$$

define

$$\mathcal{E}_r^\beta(T) \# \left\{ L \leq V : \dim L = r, \beta(\Lambda^2 L) = T \right\}.$$

Define also

$$\mathcal{F}_r^\beta(U) \# \left\{ L \leq V : \dim L = r, \beta(\Lambda^2 L) \leq U \right\}.$$

Then

$$\mathcal{F}_r^\beta(U) \sum_{T \leq U} \mathcal{E}_r^\beta(T),$$

and hence

$$\mathcal{E}_r^\beta(T) \sum_{U \leq T} \mu(U, T) \mathcal{F}_r^\beta(U),$$

where μ is the Mobius function of the subspace lattice.

The condition

$$\beta(\Lambda^2 L) \leq U$$

means that (L) is totally isotropic for the quotient map

$$\bar{\beta}_U : \Lambda^2 V \rightarrow W/U.$$

Thus the Grassmann image profile is equivalent to the collection of common isotropic-subspace counts for all quotients of the codomain.

This observer is related to, but not identical with, the tuple-based structured CSI profile. It is useful here because its two-plane collapse clarifies why level one is insufficient.

5 The full two-plane profile collapses

The case $r = 2$ of the Grassmann image profile is special.

Let

$$N_2 = \binom{\dim V}{2}_q$$

be the number of two-dimensional subspaces of (V) .

Let

$$z_\beta = \mathcal{E}_2^\beta(0)$$

and, for

$$P \in \mathbb{P}(W),$$

let

$$m_\beta(P) = \mathcal{E}_2^\beta(P).$$

Thus $m_\beta(P)$ counts two-planes whose image is the projective direction (P) .

For

$$0 \neq a \in W^*,$$

let

$$J_\beta(a) \# \{L \leq V : \dim L = 2, (a \circ \beta)|_L = 0\}.$$

If

$$H_a = \mathbb{P}(\ker a) \subseteq \mathbb{P}(W),$$

then

$$J_\beta(a) = z_\beta + \sum_{P \in H_a} m_\beta(P). \quad (5.1)$$

Let

$$d = \dim W, \quad v = \frac{q^d - 1}{q - 1}, \quad k = \frac{q^{d-1} - 1}{q - 1}.$$

Summing 5.1 over all projective covectors $[a] \in \mathbb{P}(W^*)$ gives

$$\sum_{[a]} J_\beta(a) k N_2 + (v - k) z_\beta.$$

Hence

$$z_\beta \frac{\sum_{[a]} J_\beta(a) - k N_2}{v - k}. \quad (5.2)$$

Once z_β is known, 5.1 gives the hyperplane sums of m_β . The point-hyperplane incidence matrix of projective space satisfies

$$M^T M = q^{d-2} I + \lambda J, \quad \lambda = \frac{q^{d-2} - 1}{q - 1}.$$

Thus it is invertible over $\mathbb{Q}$, and the hyperplane sums recover m_β .

It remains to observe that $J_\beta(a)$ is determined by the rank of the scalar alternating form

$$a \circ \beta.$$

Let (B) be an alternating form on an (n)-dimensional vector space, and suppose

$$\text{rank } B = 2s.$$

Then (B) has radical dimension $n - 2s$. A two-plane is nonisotropic for (B) precisely when, for one, equivalently every, ordered basis $((x, y))$, one has

$$B(x, y) \neq 0.$$

The number of ordered pairs (x, y) with $B(x, y) \neq 0$ is

$$(q^n - q^{n-2s}) q^{n-1} (q - 1).$$

Dividing by

$$|\text{GL}_2(q)|$$

gives the number of nonisotropic two-planes. Therefore

$$J(B) \binom{n}{2}_q \frac{(q^n - q^{n-2s}) q^{n-1} (q - 1)}{|\text{GL}_2(q)|}. \quad (5.3)$$

This is strictly decreasing in (s). Hence $(J(B))$ determines $\text{rank } B$, and conversely.

Theorem 5.1 (Two-plane collapse). *The full structured two-plane Grassmann image profile*

$$(z_\beta, m_\beta)$$

and the marked scalar-rank function

$$[a] \in \mathbb{P}(W^*) \longmapsto \text{rank}(a \circ \beta)$$

determine one another.

In particular, the full two-plane Grassmann image profile contains no more information than the level-one scalar-rank profile.

6 Three-plane profiles are strictly stronger

The collapse in Theorem 5.1 is special to two-planes.

We give two five-dimensional pencils over $\mathbb{F}_q$, (q) odd, with identical two-plane profiles but distinct common isotropic three-plane counts.

Let

$$W = \mathbb{F}_q^2.$$

6.1 The Kronecker pencil

Let

$$V_K = X \oplus Y, \quad \dim X = 2, \quad \dim Y = 3,$$

with bases

$$x_1, x_2$$

for (X) , and

$$y_0, y_1, y_2$$

for (Y) . Define

$$\beta_K : \Lambda^2 V_K \rightarrow W$$

by

$$\beta_K(x_1, y_0) = \beta_K(x_2, y_1) = (1, 0),$$

$$\beta_K(x_1, y_1) = \beta_K(x_2, y_2) = (0, 1),$$

and all remaining basis pairings zero.

For $(a, b) \neq (0, 0)$, the scalar form

$$a\beta_{K,0} + b\beta_{K,1}$$

has cross-block matrix

$$\begin{pmatrix} a & b & 0 & 0 & a & b \end{pmatrix}.$$

This matrix has rank 2, so the scalar alternating form has rank 4. Thus every nonzero scalar form in the pencil has rank 4.

The subspace (Y) is common totally isotropic. It is the only common totally isotropic three-space.

Indeed, suppose $L \leq V_K$ is common totally isotropic with $\dim L = 3$. Let

$$a = \dim \operatorname{pr}_X(L).$$

If $a = 0$, then $L = Y$.

If $a = 1$, then $\dim(L \cap Y) \geq 2$. But for a nonzero

$$x = \alpha x_1 + \beta x_2,$$

the map

$$Y \rightarrow W, \quad y \mapsto \beta_K(x, y)$$

has matrix

$$\begin{pmatrix} \alpha & \beta & 0 & 0 & \alpha & \beta \end{pmatrix},$$

which has kernel of dimension 1. Hence $L \cap Y$ cannot have dimension 2.

If $a = 2$, then $L \cap Y \neq 0$. But the common right radical in (Y) of (X) is zero. This is again impossible.

Therefore

$$\mathcal{E}_3^{\beta_K}(0) = 1. \tag{6.1}$$

6.2 The extension-field pencil

Let

$$E = \mathbb{F}_{q^2}$$

and let

$$V_E = \mathbb{F}_q r \oplus E^2.$$

View

$$W = E$$

as a two-dimensional $\mathbb{F}_q$ -space. Define

$$\beta_E((c, u), (c', v)) = \det_E(u, v).$$

Every nonzero scalar projection

$$E \rightarrow \mathbb{F}_q$$

composed with the determinant form is nondegenerate on E^2 and has radical exactly $\mathbb{F}_q r$. Hence every nonzero scalar form in the pencil again has rank 4.

A common totally isotropic three-space must contain (r). Its projection to E^2 must be an (E)-line. Conversely, every (E)-line gives such a three-space. Therefore

$$\mathcal{E}_3^{\beta_E}(0)|\mathbb{P}^1(E)|q^2 + 1. \quad (6.2)$$

Corollary 6.1. *The pencils β_K and β_E have identical complete two-plane image profiles but distinct three-plane image profiles:*

$$\mathcal{E}_3^{\beta_K}(0) = 1, \quad \mathcal{E}_3^{\beta_E}(0) = q^2 + 1.$$

Thus three-plane image data strictly refines two-plane image data.

7 Irreducible symplectic pencils

We now construct the main family.

Let

$$f(t) \in \mathbb{F}_q[t]$$

be monic irreducible of degree

$$m \geq 3.$$

Let

$$E = \mathbb{F}_q[t]/(f), \quad \theta = t \bmod f.$$

Let

$$V = E^2$$

viewed as a $2m$ -dimensional $\mathbb{F}_q$ -space.

Let

$$W = \mathbb{F}_q^2.$$

Define

$$\delta : E^2 \times E^2 \rightarrow E$$

by

$$\delta(x, y) = x_1 y_2 - x_2 y_1.$$

Define

$$\beta_f : V \times V \rightarrow W$$

by

$$\beta_f(x, y) (\text{Tr } E/\mathbb{F}_q \delta(x, y), \text{Tr } E/\mathbb{F}_q (\theta \delta(x, y))) . \quad (7.1)$$

Let B_0, B_1 be the two coordinate forms.

For $(a, b) \neq (0, 0)$,

$$aB_0 + bB_1 \operatorname{Tr}_{E/\mathbb{F}_q}((a + b\theta)\delta(-, -)).$$

Since $a + b\theta \neq 0$, the trace pairing is nondegenerate and δ is symplectic over (E) . Hence every nonzero scalar form in the pencil is nondegenerate over $\mathbb{F}_q$.

Consequently, β_f is radical-free and surjective.

The Pfaffian of the pencil is

$$\operatorname{Pf}(aB_0 + bB_1)c \cdot N_{E/\mathbb{F}_q}(a + b\theta) \quad (7.2)$$

for some nonzero scalar $c \in \mathbb{F}_q^\times$.

Thus the binary Pfaffian form is the homogenization of (f), up to scalar.

We use the following standard canonical-form input for alternating pencils.

Canonical-form input 7.1

Over a field of characteristic not 2, pairs of skew-symmetric matrices under congruence admit a canonical decomposition into regular and singular summands. In the regular skew-symmetric part, irreducible summands are determined by their elementary divisors. For a regular irreducible alternating pencil, the pseudo-isometry class is determined, up to projective change of pencil coordinates, by the projective equivalence class of the irreducible elementary divisor.

This is the standard Kronecker–Sergeichuk classification for pairs of skew-symmetric forms under congruence. We use it only in this regular irreducible case.

8 Coordinate form of CSI for pencils

Identify

$$W^*$$

with the two-dimensional space of linear polynomials

$$\mathcal{L} = a + bt : a, b \in \mathbb{F}_q.$$

Let

$$U \leq (W^*)^r.$$

Choose a basis of (U) . We represent it by an $s \times r$ matrix

$$C_U(t)$$

whose entries are linear polynomials in (t) .

Let (A) denote multiplication by θ on $V = E^2$. Using the nondegenerate form B_0 to identify (V) with V^* , the map defining

$$\rho_r^{\beta_f}(U)$$

is represented by

$$C_U(A) : V^r \rightarrow V^s.$$

Therefore

$$\rho_r^{\beta_f}(U) \operatorname{rank}_{\mathbb{F}_q} C_U(A). \quad (8.1)$$

More explicitly, if

$$C_U(t) = (c_{ij}(t)),$$

then $C_U(A)$ is the block map whose $((i,j))$ -block is multiplication by

$$c_{ij}(\theta)$$

on E^2 .

This formula is the bridge between support-image profiles and polynomial matrix rank.

9 Complete blindness below degree (m)

Lemma 9.1. *Let (f) be irreducible of degree (m) . If*

$$r < m,$$

then

$$\rho_r^{\beta_f}(U) 2m \cdot \operatorname{rank}_{\mathbb{F}_q(t)} C_U(t). \quad (9.1)$$

In particular, the value is independent of (f) .

Proof. Let

$$k = \operatorname{rank}_{\mathbb{F}_q(t)} C_U(t).$$

Take the Smith normal form of $C_U(t)$ over $\mathbb{F}_q[t]$:

$$P(t)C_U(t)Q(t) \operatorname{diag}(s_1(t), \dots, s_k(t), 0, \dots, 0),$$

where $(P(t), Q(t))$ are unimodular.

The product

$$s_1(t) \cdots s_k(t)$$

divides every nonzero $k \times k$ minor. Since the entries of $C_U(t)$ have degree at most 1, every such minor has degree at most $k \leq r$. Hence

$$\deg s_i(t) < m$$

for each nonzero invariant factor s_i .

Because (f) is irreducible of degree (m) , every nonzero $s_i(t)$ is coprime to $(f(t))$. Therefore

$$s_i(A)$$

is invertible on $V = E^2$.

The evaluated matrices

$$P(A), Q(A)$$

are invertible because (P,Q) are unimodular.

Thus $C_U(A)$ has exactly (k) invertible (V)-blocks and rank

$$k \dim_{\mathbb{F}_q} V = 2mk.$$

□

We now verify the projective invariance used later.

□

Lemma 9.2 (Projective invariance below threshold). *Let*

$$g \in PGL_2(q).$$

For every $U \leq (W^)^r$,*

$$\text{rank}_{\mathbb{F}_q(t)} C_{gU}(t) = \text{rank}_{\mathbb{F}_q(t)} C_U(t).$$

Consequently, if $r < m$, then

$$\rho_r^{\beta_f}(gU) = \rho_r^{\beta_f}(U).$$

Proof. Choose a representative

$$g(t) = \frac{\alpha t + \beta}{\gamma t + \delta}.$$

The induced action on linear pencil coordinates sends a linear form in (t) to a nonzero scalar multiple of

$$(\gamma t + \delta)\ell(g(t)).$$

Thus, after clearing a common denominator, $C_{gU}(t)$ is obtained from $C_U(t)$ by:

□

- applying the field automorphism of $\mathbb{F}_q(t)$ induced by the fractional-linear substitution $t \mapsto g(t)$;
- multiplying rows or columns by nonzero rational functions;
- changing bases in (U) and in $(W^*)^r$, i.e. applying invertible row and column operations over $\mathbb{F}_q$.

All these operations preserve rank over the rational function field $\mathbb{F}_q(t)$. Therefore

$$\text{rank}_{\mathbb{F}_q(t)} C_{gU}(t) = \text{rank}_{\mathbb{F}_q(t)} C_U(t).$$

The final statement follows from Lemma 9.1.

□

Theorem 9.3 (Maximal low-level symmetry). *For every monic irreducible (f) of degree (m),*

$$\Gamma_{\leq r}(\beta_f) = PGL_2(q) \quad \text{for every } r < m. \quad (9.2)$$

Moreover, if (f,h) are any two monic irreducible polynomials of degree (m), then

$$\text{CSI}_{\leq m-1}(\beta_f) \text{CSI}_{\leq m-1}(\beta_h). \quad (9.3)$$

Thus all support-image observers of arity below (m) are completely blind to the polynomial (f).

Proof. For $r < m$, Lemma 9.1 gives

$$\rho_r^{\beta_f}(U) 2m \cdot \text{rank}_{\mathbb{F}_q(t)} C_U(t),$$

which is independent of (f). This proves 9.3.

Lemma 9.2 shows that every projective change of pencil coordinates preserves the rank transform through level $r < m$. Hence

$$PGL_2(q) \leq \Gamma_{\leq r}(\beta_f).$$

The reverse inclusion is automatic from the definition, since $\Gamma_{\leq r}(\beta_f)$ is a subgroup of $PGL(W) = PGL_2(q)$. Therefore

$$\Gamma_{\leq r}(\beta_f) = PGL_2(q).$$

□

□

10 Level (m) detects the polynomial

Let C_f be the companion matrix of (f), and define the $m \times m$ linear matrix polynomial

$$L_f(t) = tI_m - C_f.$$

Then

$$\det L_f(t) = f(t).$$

Let

$$U_f \leq (W^*)^m$$

be the row space of $L_f(t)$.

For the pencil β_f , evaluation at (A) gives

$$L_f(A),$$

which is singular because

$$f(A) = 0.$$

Thus

$$\rho_m^{\beta_f}(U_f) < 2m^2. \quad (10.1)$$

Now let (h) be another monic irreducible polynomial of degree (m). Let A_h be multiplication by a root of (h) on E_h^2 .

For

$$g \in PGL_2(q),$$

the transformed subspace

$$gU_f$$

is represented by a linear matrix polynomial whose determinant is, up to a nonzero scalar, the projective transform

$$g \cdot f.$$

If (h) is not projectively equivalent to (f), then (h) and $g \cdot f$ are distinct irreducibles of degree (m), hence coprime. Therefore

$$(g \cdot f)(A_h)$$

is invertible, and the evaluated linear matrix is invertible. Hence

$$\rho_m^{\beta_h}(gU_f) = 2m^2. \quad (10.2)$$

Equations 10.1 and 10.2 give a level-(m) witness separating β_f from every projectively inequivalent β_h .

Theorem 10.1 (Exact level-(m) classification). *Let $f, h \in \mathbb{F}_q[t]$ be monic irreducible polynomials of degree $m \geq 3$. Then*

$$\text{CSI}_{\leq m}(\beta_f) \text{CSI}_{\leq m}(\beta_h)$$

up to the natural $PGL_2(q)$ -action if and only if

$$h \in PGL_2(q) \cdot f.$$

Equivalently, β_f and β_h are pseudo-isometric as alternating pencils.

Proof. If h is not projectively equivalent to f , the companion linearization above gives an explicit $U_f \leq (W^*)^m$ such that the rank value for β_f is singular, while every projective translate has full rank for β_h . Thus the level- m CSI profiles differ.

Conversely, if h is projectively equivalent to f , then the corresponding Pfaffian binary forms are projectively equivalent. By the canonical-form input for regular irreducible skew-symmetric pencils, the pencils β_f and β_h are pseudo-isometric. Hence all CSI profiles agree.

□

Combining Theorems 9.3 and 10.1, the exact CSI separation depth of the irreducible degree-(m) pencil family is

$$\boxed{m.}$$

□

11 True and false projective symmetry

Let

$$\text{Aut}_W(\beta_f) \leq PGL_2(q)$$

be the projective image of the pseudo-isometry group of β_f .

The Pfaffian form implies

$$\text{Aut}_W(\beta_f) \leq \text{Stab}_{PGL_2(q)}(f).$$

The converse follows from the regular symplectic pencil classification: a projective transformation stabilizing the elementary divisor is realized by a pseudo-isometry of the pencil. Hence

$$\text{Aut}_W(\beta_f) \text{Stab}_{PGL_2(q)}(f). \quad (11.1)$$

From Theorem 9.3,

$$\Gamma_{\leq r}(\beta_f) = PGL_2(q) \quad (r < m).$$

From Theorem 10.1,

$$\Gamma_{\leq m}(\beta_f) = \text{Stab}_{PGL_2(q)}(f). \quad (11.2)$$

Therefore support-image observers exhibit maximal false symmetry below the threshold:

$$\boxed{\Gamma_{\leq r}(\beta_f) = PGL_2(q) \quad (r < m),}$$

but at the threshold the false symmetry collapses exactly to the true projective automorphism group:

$$\boxed{\Gamma_{\leq m}(\beta_f) = \text{Aut}_W(\beta_f).}$$

12 Direct indecomposability of the associated groups

Assume in this section that

$$q = p$$

is an odd prime. Then G_{β_f} is a finite (p)-group of class two and exponent (p).

Since β_f is radical-free and surjective,

$$Z(G_{\beta_f}) = G'_{\beta_f} = W.$$

Thus G_{β_f} is special.

Proposition 12.1. *The group G_{β_f} is directly indecomposable.*

Proof. It is enough to show that the alternating pencil admits no nontrivial orthogonal direct decomposition.

Suppose

$$V = V_1 \perp V_2$$

is a nontrivial decomposition orthogonal for both B_0 and B_1 .
Since B_0 is nondegenerate, each V_i is B_0 -nondegenerate.
Let (A) be multiplication by θ on $V = E^2$. Then

$$B_1(x, y) = B_0(Ax, y).$$

Orthogonality for B_1 implies that each V_i is (A)-invariant. Since (A) has irreducible minimal polynomial (f), (A)-invariant $\mathbb{F}_q$ -subspaces of E^2 are precisely (E)-subspaces.

A proper nonzero (E)-subspace of E^2 is an (E)-line. Every (E)-line is totally isotropic for

$$\delta(x, y) = x_1y_2 - x_2y_1,$$

and hence for B_0 . This contradicts the B_0 -nondegeneracy of V_i .

Therefore no nontrivial orthogonal direct decomposition exists, and G_{β_f} is directly indecomposable.

□

The order of the group is

$$|G_{\beta_f}| = p^{2m+2}.$$

Thus the exact threshold can be written as

$$m = \frac{\log_p |G_{\beta_f}| - 2}{2}.$$

□

13 Exponentially large low-level fibres

Let

$$I_q(m)$$

denote the number of monic irreducible polynomials of degree (m) over $\mathbb{F}_q$. Then

$$I_q(m) \frac{1}{m} \sum_{e|m} \mu(e) q^{m/e} \frac{q^m}{m} + O_q(q^{m/2}).$$

The group $PGL_2(q)$ has order

$$|PGL_2(q)| = q(q^2 - 1).$$

Therefore the number of $PGL_2(q)$ -orbits of irreducible degree-(m) polynomials is at least

$$\frac{I_q(m)}{|PGL_2(q)|} \frac{q^m}{m, q(q^2 - 1)} + O_q(q^{m/2}). \quad (13.1)$$

Choosing one representative from each orbit gives a family

$$\mathcal{F}_m$$

of pairwise non-pseudo-isometric irreducible pencils.

For $q = p$ prime, these yield pairwise nonisomorphic special directly indecomposable class-two exponent-(p) groups.

Every member of $\mathcal{F}_m$ has the same CSI profile through level

$$m - 1.$$

At level (m), the profile separates the family.

Thus a single $\text{CSI}_{\leq m-1}$ -fibre contains

$$q^{m-o(m)}$$

pairwise nonisomorphic objects. In group order,

$$q^m = q^{(\log_q |G| - 2)/2} = |G|^{1/2} q^{-1}.$$

So the low-level fibre has size

$$|G|^{1/2-o(1)}.$$

14 Generic triviality at the threshold

The threshold often destroys all projective symmetry.

Proposition 14.1. *Suppose*

$$\gcd(m, |PGL_2(q)|) = 1.$$

Then every monic irreducible polynomial of degree (m) has trivial $PGL_2(q)$ -stabilizer.

Proof. Let α be a root of (f). The roots of (f) are the Frobenius orbit

$$\alpha, \alpha^q, \dots, \alpha^{q^{m-1}}.$$

If

$$1 \neq g \in PGL_2(q)$$

stabilizes (f), then (g) permutes this Frobenius orbit. Hence for some (j),

$$g\alpha = \alpha^{q^j}.$$

The induced permutation on the orbit has order dividing (m). On the other hand, the order of (g) divides

$$|PGL_2(q)|.$$

Since

$$\gcd(m, |PGL_2(q)|) = 1,$$

the induced permutation must be trivial. Thus (g) fixes α . But a nonidentity projective linear transformation over $\mathbb{F}_q$ has at most two fixed points over the algebraic closure, and cannot fix a degree-(m) element with $m \geq 3$. Therefore $g = 1$.

□

There are infinitely many such (m). For these degrees,

$$\Gamma_{\leq m-1}(\beta_f) = PGL_2(q), \quad \Gamma_{\leq m}(\beta_f) = 1.$$

A probabilistic version also holds. □

Proposition 14.2. *Let (f) be uniformly random among monic irreducible polynomials of degree (m). Then*

$$\Pr \left(\text{Stab}_{PGL_2(q)}(f) \neq 1 \right) q^{-\Omega_q(m)}.$$

More explicitly,

$$\Pr \left(\text{Stab}_{PGL_2(q)}(f) \neq 1 \right) O_q \left(q^{-m/(q+1)} \right).$$

Proof. It is equivalent to sample a degree-(m) element $\alpha \in \overline{\mathbb{F}}_q$, up to Frobenius orbit.

Fix

$$1 \neq g \in PGL_2(q)$$

of order (h). Every element of $PGL_2(q)$ has order at most $q + 1$, after including the unipotent case since $p \leq q < q + 1$.

If (g) stabilizes the Frobenius orbit of a degree-(m) element α , then

$$g\alpha = \alpha^{q^j}$$

for some $0 \leq j < m$.

If $j = 0$, then (g) fixes α . A nonidentity projective linear transformation has at most two fixed points over the algebraic closure, so this cannot occur for a degree-(m) element with $m \geq 3$.

Thus $1 \leq j < m$. Applying (g) repeatedly gives

$$\alpha = g^h \alpha = \alpha^{q^{jh}},$$

so

$$m \mid jh.$$

Hence (j) is a multiple of $m/\gcd(m, h)$, and there are at most $h - 1 \leq q$ possible values of (j). For fixed (j), the equation

$$g(x) = x^{q^j}$$

has at most $q^j + 1$ solutions in the algebraic closure. The largest possible (j) is at most

$$m \left(1 - \frac{1}{h} \right) \leq m \left(1 - \frac{1}{q+1} \right).$$

Thus the number of degree-(m) elements whose Frobenius orbit is stabilized by this fixed (g) is

$$O_q \left(q^{m(1-1/(q+1))} \right).$$

The number of degree-(m) elements over $\mathbb{F}_q$ is

$$q^m + O_q(q^{m/2}).$$

Therefore, for fixed $g \neq 1$, the probability that (g) stabilizes the orbit is

$$O_q\left(q^{-m/(q+1)}\right).$$

Taking a union bound over the finite group $PGL_2(q)$ gives the claim.

□

Thus a random member of the irreducible pencil family has trivial true projective automorphism group with exponentially high probability, while its support-image observer symmetry is still the full $PGL_2(q)$ below level (m) .

□

15 Main theorem

We collect the results.

Theorem 15.1 (Sharp support-image threshold for irreducible alternating pencils). *Let (q) be odd and $m \geq 3$. For every monic irreducible polynomial $f \in \mathbb{F}_q[t]$ of degree (m) , let*

$$\beta_f : \Lambda^2 \mathbb{F}_q^{2m} \rightarrow \mathbb{F}_q^2$$

be the alternating pencil defined in 7.1.

Then:

- For every $r < m$, the structured support-image profile $\text{CSI}_{\leq r}(\beta_f)$ is independent of (f) .
- For every $r < m$,

$$\Gamma_{\leq r}(\beta_f) = PGL_2(q).$$

- At level (m) , the profile detects (f) up to the natural $PGL_2(q)$ -action.
- Hence the exact CSI separation depth of the irreducible degree- (m) pencil family is (m) .
- A single $\text{CSI}_{\leq m-1}$ -fibre contains at least

$$\frac{q^m}{m, q(q^2 - 1)} + O_q(q^{m/2})$$

distinct pseudo-isometry classes.

- For a proportion

$$1 - O_q(q^{-m/(q+1)})$$

of irreducible (f), the level-(m) residual observer symmetry is trivial:

$$\Gamma_{\leq m}(\beta_f) = 1.$$

If $q = p$ is prime, the corresponding Baer groups G_{β_f} are directly indecomposable special class-two exponent-(p) groups of order

$$p^{2m+2},$$

and the exact observer threshold is

$$m = \frac{\log_p |G_{\beta_f}| - 2}{2}.$$

Proof. Parts 1 and 2 are Theorem 9.3. Parts 3 and 4 are Theorem 10.1. Part 5 is the orbit count in Section 13. Part 6 is Proposition 14.2 together with 11.2. The group-theoretic statement follows from Proposition 12.1 and the order calculation.

□

□

16 Consequences for star, flag, and rank-support profiles

The support-image profile dominates the rank-support and flag profiles.

For

$$\mathbf{x} \in V^r,$$

the rank of

$$A_{\mathbf{x}} : V \rightarrow W^r$$

is the dimension of the support image

$$\text{im } A_{\mathbf{x}}.$$

Thus CSI determines the rank distribution

$$N_r^\beta(a) \# \{\mathbf{x} \in V^r : \text{rank } A_{\mathbf{x}} = a\}.$$

It also determines the aggregated flag profile, because for

$$S \subseteq 1, \dots, r,$$

the rank of the coordinate subtuple is the dimension of the projection of

$$\text{im } A_{\mathbf{x}} \leq W^r$$

to the coordinates in (S).

Therefore Theorem 15.1 implies:

Corollary 16.1. *For the family*

$$\beta_f : \deg f = m,$$

all star profiles, rank-support profiles, and aggregated flag profiles of arity

$$r < m$$

are identical.

At arity (m) , structured CSI separates the irreducible-pencil family exactly.

This shows that the obstruction is not merely a failure of crude commuting probabilities; it persists through the full structured support-image hierarchy below the threshold.

17 Interpretation

The theorem identifies a precise mechanism behind support-image depth.

For an irreducible degree- (m) pencil, all polynomial tests of matrix size below (m) see only generic rank over the rational function field

$$\mathbb{F}_q(t).$$

The irreducible polynomial (f) cannot divide any nonzero determinant of degree $< m$. Thus every low-level observer is blind to (f) .

At level (m) , a determinant of degree (m) can equal (f) . The companion linearization

$$tI_m - C_f$$

is the minimal witness. Its determinant is precisely (f) , and it becomes singular exactly on the corresponding pencil.

Thus the observer transition is governed by the first possible degree of a Pfaffian/eigenvalue witness:

support-image depthdegree of the irreducible elementary divisor.
--

Below threshold, the observer sees only rational-function-field rank. At threshold, it sees the finite-field point θ .

18 Further directions

18.1 Higher-dimensional regular spaces

The present paper treats pencils, i.e. $\dim W = 2$. The natural next case is a regular alternating space of derived rank $d \geq 3$. One expects similar thresholds governed by the minimal degree of determinantal or Pfaffian equations cutting out the spectral data.

18.2 Generic support-image rigidity

The result suggests that generic random alternating spaces may have an observer threshold governed by the minimal degree of their projective degeneracy locus. Proving this would require anti-concentration for random Pfaffian systems.

18.3 Deterministic Grassmannian projection asymmetry

The full deterministic two-plane profile collapses to scalar ranks, but higher-plane image profiles may define asymmetric projective fibre counts for random projections of Grassmannians. This remains an attractive route to generic deterministic support-image canonization.

18.4 Algorithms

The level-(m) witness is explicit: a companion linearization. This suggests an algorithmic strategy for regular pencils: search for the first arity (r) at which a support-image rank defect appears. The first defect reveals the degree of the irreducible elementary divisor.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, ChatGPT, by OpenAI, was used to assist with mathematical drafting, formalization, review, and editing. This work is shared as a preliminary AI-assisted mathematical note. The mathematical content may have been only partially reviewed and may contain errors; it should not be treated as peer-reviewed or as a fully verified manuscript.

References

- [1] V. A. Bovdi, T. G. Gerasimova, M. A. Salim, and V. V. Sergeichuk. Reduction of a pair of skew-symmetric matrices to its canonical form under congruence. *Linear Algebra and its Applications* 498 2016, 403–433.
- [2] R. Gow and G. McGuire. Invariant rational functions, linear fractional transformations and irreducible polynomials over finite fields. *Finite Fields and Their Applications* 78 2022, 101977.
- [3] R. A. Horn and V. V. Sergeichuk. Canonical forms for complex matrix congruence and $(-)$ -congruence*. *Linear Algebra and its Applications* 416 2006, 1010–1032.
- [4] R. Lidl and H. Niederreiter. *Finite Fields*. Cambridge University Press, 1997.
- [5] L. Reis. Invariant theory of a special group action on irreducible polynomials over finite fields. *Designs, Codes and Cryptography* 87 2019, 1095–1117.
- [6] V. V. Sergeichuk. Canonical matrices of bilinear and sesquilinear forms. *Linear Algebra and its Applications* 343–344 2002, 43–60.
- [7] V. V. Sergeichuk. Canonical matrices of forms and pairs of forms over finite and (p) -adic fields. *Linear Algebra and its Applications* 438 2013, 2322–2339.
- [8] J. B. Wilson. Decomposing (p) -groups via Jordan algebras. *Journal of Algebra* 322 2009, 2642–2679.
- [9] P. A. Brooksbank and J. B. Wilson. Computing isometry groups of Hermitian maps. *Transactions of the American Mathematical Society* 364 2012, 1975–1996.
- [10] Y. Li and Y. Qiao. Linear algebraic analogues of the graph isomorphism problem and the Erdos–Renyi model. *SIAM Journal on Computing* 51 2022, 926–963.